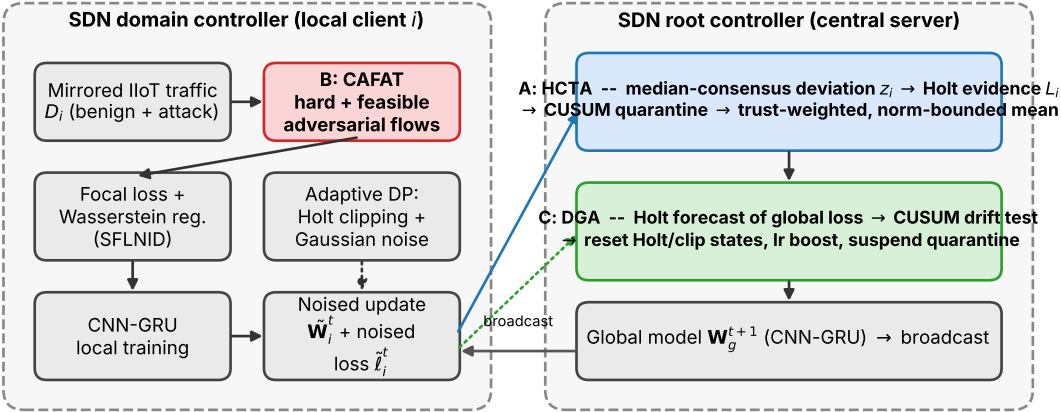
SDN domain controller (local client i)
Mirrored IIoT traffic D_i (benign + attack)
B: CAFAT hard + feasible adversarial flows
Focal loss + Wasserstein reg. (SFLNID)
Adaptive DP: Holt clipping + Gaussian noise
CNN-GRU local training
Noised update $\tilde{\mathbf{W}}_i^t$ + noised loss $\tilde{\ell}_i^t$
SDN root controller (central server)
A: HCTA -- median-consensus deviation z_i → Holt evidence L_i → CUSUM quarantine → trust-weighted, norm-bounded mean
C: DGA -- Holt forecast of global loss → CUSUM drift test reset Holt/clip states, lr boost, suspend quarantine
broadcast
Global model $\mathbf{W}_g^{t+1}$ (CNN-GRU) → broadcast
Malicious controllers: poisoning (training) | Evasion adversary: perturbs controllable features (inference) | Drift: new attack variants